

G001 Hydraulic System Reference Manual

System Overview and Components

The G001 hydraulic system employs a triple-redundant architecture consisting of three independent hydraulic circuits designated System A, System B, and System C. Each system operates at a nominal pressure of 3000 PSI $\pm$ 150 PSI and utilizes MIL-PRF-83282 Type II synthetic hydraulic fluid with a total system capacity of 42 gallons distributed across three separate reservoirs.

System A is powered by an engine-driven pump (EDP) mounted on the left engine accessory gearbox, supplemented by an electric motor-driven pump (EMDP-1) for ground operations and emergency backup. System B derives primary power from the right engine EDP with EMDP-2 providing redundancy. System C operates exclusively through two electric motor-driven pumps (EMDP-3 and EMDP-4) and serves as the emergency backup system, drawing power from the essential AC bus during normal operations and automatically switching to battery power during total electrical failure.

Each reservoir incorporates a pressurization system utilizing bleed air at 45 PSI to ensure positive fluid flow to pump inlets and prevent cavitation at altitude. Low-level sensors trigger amber EICAS caution messages at 0.65 reservoir capacity, while critically low levels (0.35 capacity) generate red warning messages and automatic system isolation.

Primary hydraulic system components include:

- Return filters with bypass valves and differential pressure indicators
- Thermal relief valves set at 3600 PSI
- Priority valves ensuring flight control authority during low-pressure conditions
- Accumulator assemblies maintaining system pressure during transient demands
- Fire shutoff valves (FSOV) for each EDP, controlled via overhead panel switches

The hydraulic distribution manifold routes pressurized fluid to the following subsystems:

1. Primary and secondary flight control actuators
2. Landing gear extension/retraction and steering mechanisms
3. Wheel brake system and anti-skid components
4. Thrust reverser actuation units
5. Flap and slat drive mechanisms

6. Cargo door actuators and latching systems

Cross-feed capability between System A and System B exists through electrically-controlled isolation valves, enabling single-system operation of critical flight controls during emergency scenarios. The hydraulic system BIT routine executes automatically during preflight power-up, verifying pump operation, pressure transducer accuracy, and valve functionality within 90 seconds.

Hydraulic Pressure Normal Operating Ranges

The G001 hydraulic system operates with three independent hydraulic circuits designated as System A, System B, and System C. Each system maintains nominal operating pressure of 3000 PSI $\pm$ 150 PSI during normal flight operations. The hydraulic pressure indication is displayed on the primary EICAS hydraulic synoptic page, with digital readouts and analog bar graphs providing real-time monitoring capability.

Normal Pressure Parameters:

- **System A (Green):** 2850-3150 PSI nominal, 2700 PSI minimum acceptable
- **System B (Blue):** 2850-3150 PSI nominal, 2700 PSI minimum acceptable
- **System C (Yellow):** 2850-3150 PSI nominal, 2700 PSI minimum acceptable
- **Accumulator Precharge:** 1450-1550 PSI (nitrogen)
- **Low Pressure Caution:** 2400 PSI (amber EICAS message)
- **Critical Low Pressure Warning:** 1800 PSI (red EICAS message)

During engine start and APU operation, hydraulic pressure may temporarily fluctuate between 2600-2900 PSI as demand pumps cycle and system components initialize. This is considered normal behavior during the first 30 seconds of pump operation. The engine-driven pumps (EDP) for Systems A and B typically stabilize within 15 seconds of engine N2 reaching 50%.

System C, powered by electric motor-driven pumps (EMDP), achieves full pressure within 8-10 seconds of pump activation. When operating on electric pumps alone (ground operations or single-engine taxi), expect pressure readings approximately 100 PSI lower than EDP-supplied systems due to reduced flow capacity.

Pressure Verification Procedure:

1. Select HYD synoptic page on EICAS display
2. Verify all three system pressures indicate green arc (2850-3150 PSI)
3. Check quantity indicators show 90-100% (RF level)

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4. Confirm pump status indicators display ON with no fault annunciations
5. Monitor for pressure decay—acceptable rate is less than 50 PSI per minute with pumps off

Pressure transients exceeding 3400 PSI trigger overpressure relief valves. Sustained operation above 3300 PSI generates a BIT fault code HYD-421 and requires maintenance investigation of system relief valve calibration.

Common Fault Codes and Indications

The G001 hydraulic system incorporates comprehensive Built-In Test (BIT) capabilities through the Hydraulic System Control Unit (HSCU), which continuously monitors system parameters and generates fault codes when anomalies are detected. These codes are displayed on the Engine Indication and Crew Alerting System (EICAS) or Electronic Centralized Aircraft Monitor (ECAM), depending on aircraft configuration.

Primary Fault Code Categories:

- **HYD-100 Series:** Pressure-related faults including low system pressure, pump cavitation, or accumulator precharge loss
- **HYD-200 Series:** Temperature exceedances in reservoir, return lines, or pump case drain systems
- **HYD-300 Series:** Fluid quantity issues, including low level warnings and reservoir overfill conditions
- **HYD-400 Series:** Electrical faults affecting pump motor controllers, solenoid valves, or sensor circuits
- **HYD-500 Series:** Control system faults within the HSCU or communication bus errors

Common fault code HYD-142 indicates System A pressure below 2,850 PSI for more than 3 seconds during normal operation. This typically results from Engine-Driven Pump (EDP) failure, demand pump malfunction, or significant system leakage. The HSCU will automatically command the associated Electric Motor-Driven Pump (EMDP) online if available.

Fault code HYD-217 signals hydraulic fluid temperature exceeding 95°C (203°F) in the return manifold. This condition triggers automatic pump shutdown to prevent thermal damage and fluid degradation. Operators should verify adequate cooling airflow and inspect for heat exchanger blockage or bypass valve malfunction.

Fault Code Retrieval Procedure:

1. Access Maintenance page on Multifunction Display (MFD)
2. Select HYDRAULIC SYSTEM submenu
3. Navigate to FAULT HISTORY tab
4. Review active and stored fault codes with associated timestamps
5. Record fault occurrence count and flight phase data
6. Cross-reference codes with AMM Chapter 29 for troubleshooting procedures

Intermittent codes appearing fewer than three times per 100 flight hours typically indicate transient sensor noise rather than hard failures. Persistent or recurring codes require immediate investigation per approved maintenance practices.

Actuator Seal Inspection Procedure

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Hydraulic actuator seal integrity is critical to maintaining system pressure and preventing contamination ingress that could compromise flight control surface operation. This inspection procedure applies to all primary and secondary flight control actuators within the G001 simulator's hydraulic system architecture, including aileron, elevator, rudder, spoiler, and flap actuators.

Prior to commencing seal inspection, ensure all three hydraulic systems (Green, Blue, and Yellow) are depressurized and the electric hydraulic pump (ELEC HYD PUMP) switches are positioned to OFF. Verify hydraulic quantity indications on ECAM SD page read below 10% or that manual drain procedures have been completed per Section 4.2.3.

Inspection Sequence:

1. Remove actuator protective covers and clean external surfaces using approved MIL-H-83282 compatible solvent
2. Visually inspect rod seals for evidence of weeping, scoring, or discoloration indicating thermal degradation
3. Check wiper seal condition for cuts, tears, or compression set exceeding 0.015 inches
4. Examine piston seal grooves using borescope through service ports (where applicable)
5. Measure rod straightness using dial indicator; maximum permissible runout is 0.003 inches per foot
6. Inspect backup rings for extrusion damage or cracking

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7. Verify proper installation of seal retaining hardware with calibrated torque wrench (45-55 in-lbs)

Document all findings using Maintenance Action Form MAF-G001-HYD-ACT. Any seal exhibiting fluid bypass, visible damage, or dimensional deviation beyond specified tolerances requires immediate replacement using approved seal kit part numbers listed in Component Maintenance Manual Chapter 29-40-00.

Following seal replacement, conduct operational leak check at normal system pressure (3000 PSI $\pm$ 150 PSI) for minimum 15 minutes. Monitor ECAM HYD page for pressure decay exceeding 50 PSI per minute, which indicates seal installation deficiency or component damage requiring further investigation.

Troubleshooting Hydraulic Pressure Drops

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Hydraulic pressure drops in the G001 system typically manifest through EICAS caution messages HYD SYS 1 PRESS, HYD SYS 2 PRESS, or HYD SYS 3 PRESS, accompanied by amber indications on the hydraulic synoptic display. Normal operating pressure ranges from 2,850 to 3,100 PSI for all three systems. Pressure drops below 2,400 PSI will trigger caution alerts, while drops below 1,800 PSI generate warning messages and may result in automatic system isolation.

When investigating pressure anomalies, begin with the hydraulic synoptic page on the MFD to identify which system is affected. Cross-reference fluid quantity indicators, as low reservoir levels often precede pressure drops. The system incorporates pressure transducers (PT-101, PT-201, PT-301) that feed data to the hydraulic control unit (HCU) and can be interrogated through the maintenance diagnostic system.

Initial Troubleshooting Procedure:

1. Verify hydraulic pump operation on affected system - check pump ON light illuminated
2. Monitor fluid quantity - minimum 0.85 gallons required for each system
3. Check engine-driven pump (EDP) or electric motor-driven pump (EMDP) circuit breakers on overhead panel
4. Review maintenance pages for fault codes H-2401 through H-2499 (pressure-related faults)
5. Inspect hydraulic synoptic for pump cavitation indicators or overheat warnings

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6. Verify PTU (Power Transfer Unit) not inadvertently activated, causing pressure transfer between systems
7. Check accumulator precharge pressure using ground service panel - should read 1,000 PSI ± 50

Common causes include internal pump wear, filter bypass valve activation due to contamination, or leakage past actuator seals. Transient pressure drops during high-demand operations (gear extension, flap deployment) may indicate marginal pump performance. The BIT function can be activated through the maintenance panel to cycle each pump independently and monitor pressure response curves. Fault code H-2415 specifically indicates pressure sensor drift, requiring recalibration or replacement of the affected transducer before further troubleshooting.